

Pablo Leyva

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EDUCATION

New Jersey Institute of Technology

B.S. in Applied Mathematics (Statistics) & Computer Science (AI Concentration)

Newark, NJ

Aug 2023 – May 2027

- **Relevant Coursework:** Reinforcement Learning, Deep Learning, Probability, Linear Algebra, Convex Optimization

EXPERIENCE

Apple

May 2026 – July 2026

Enterprise GenAI Product Research Intern

Cupertino, CA

- Building **model-based tools** to curate and score **RL post-training data** for a continually-learning SLM, studying how **training data quality** drives **out-of-distribution generalization** on agentic Apple Wallet and Apple Pay workflows
- Designing **RL environments with verifiable rewards** built on **static program analysis** (AST-level correctness oracles for generated payment-flow code); implementing **GRPO** and **PPO** on-policy training with **vLLM** rollouts and **reward model** training over preference data

Undergrad Research and Innovation (URI) - Agentic Reasoning SLM

Sep 2025 – Present

Research Fellow

Newark, NJ

- Training a **3B-parameter reasoning SLM** via **SFT** → **DPO** → **RL fine-tuning (GRPO)** on math-proof and logic-puzzle datasets; targeting a **public preprint** on one-step reasoning generalization in small language models
- Constructed **synthetic RL environment** with verifiable reward signals for mathematical proof search; ran **ablation studies** across reward shaping strategies and data-mixing ratios to isolate sources of reasoning gain
- Ran **mechanistic interpretability** probes and **safety evaluations** (refusal consistency, reward hacking, out-of-distribution robustness) designing and maintaining benchmarks for the SLM against larger LLM baselines on **lm-eval-harness**

Apple

May 2025 – August 2025

AI/ML Product Engineering Intern

Cupertino, CA

- Built an **agentic AI benchmark suite** evaluating **multi-agent trajectories** (tool-use success, groundedness, hallucination rate) against Claude and GPT baselines; results directly informed **reward signals** and fine-tuning priorities
- Architected **multi-agent orchestration system** integrating the Apple Foundation Model (AFM) with Model Context Protocol (MCP); collected **trajectory-level preference data** across 10k+ rollouts to bootstrap a **reward model prototype** for downstream RLHF
- Designed **tool selection and execution** with LangGraph using **TouchID Human-in-the-Loop validation**; presented prototype to senior leadership as evidence for alignment-aware agentic payment flows

Caterpillar Inc.

May 2024 – August 2024

Software Engineering Intern

Chicago, IL

- Built a **multi-agent orchestration** system that automated the **end-to-end SDLC** for a team developing **ML algorithms**, coordinating specialized agents to **generate unit tests**, triage **CI build/test failures** from logs, and draft **PR reviews**, closing the loop with **execution-based validation** (run tests → parse results → fix) before proposing a merge, freeing researchers to focus on algorithm work
- Engineered the full **agent stack** with **LangGraph** and **LangChain**: a **ReAct-style planner** decomposing tickets into steps, a **tool registry** exposed over **MCP (Model Context Protocol)** (test runner, git, log search, code search) with **tool-calling/function calling**, **memory** over the repo and prior runs, and a stateful **orchestration loop** with retry and fallback routing
- Implemented **guardrails** for safe autonomy: **sandboxed code execution**, output schema validation, **regression and diff-size checks**, and **human-in-the-loop** approval before any commit, plus structured **tracing** of every tool call and token cost for full observability

PROJECTS

Self-Extending Agent for Automated Code Generation & Testing | PyTorch, TRL, Gymnasium, vLLM, LoRA, Weights & Biases

- Built an **agent harness** that generates, executes, and self-debuges **Python tools** from natural-language specs in a closed **generate** → **test** → **validate** → **persist** loop, growing a compounding library of verified tools; curates the resulting **on-policy trajectories** as **RL training data** to **fine-tune a per-user policy (LoRA)** with **GRPO** against **verifiable rewards**, cutting per-turn tokens by ~60%

COMMUNITY OUTREACH

President, Vice President - External Affairs, Event Coordinator

October 2023 – Present

Society of Hispanic Professional Engineers (SHPE)

Newark, NJ

- Leading a team of 20 to organize professional events for 300+ students; grew external company partnerships 6x and averaged 200% growth in weekly member engagement, launched an the first iOS app in the university's history in 27 days

TECHNICAL SKILLS

Languages : Python (fluent), C++, CUDA (working), Java

ML Frameworks : PyTorch, JAX, Hugging Face Transformers, TRL, DeepSpeed, FSDP, vLLM, Ray

Inference & Serving : vLLM, FastAPI microservices, sub-100ms latency serving (10K+ req/day)

Reinforcement Learning : PPO, GRPO, DPO, RLHF, reward modeling, Gymnasium, RL environment design, preference data

LLM Training & Evaluation : SFT, LoRA/QLoRA, fine-tuning, distillation, synthetic data, lm-eval-harness, safety evaluations, ablation studies

Agentic : LangGraph, LangChain, MCP (Model Context Protocol), tool-calling/function calling, multi-agent orchestration, ReAct, planning, memory, guardrails, human-in-the-loop